

How We Recovered the Digit

A Plain-English Lesson on the ERT Project —

read it start to finish, nothing skipped

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This guide is written so that anyone in the group can *understand* the project well enough to explain it to someone else. We do not assume you remember the lectures. Every new word is defined the moment it appears, and every idea is built on the one before it. It is long on purpose: we would rather over-explain than leave you stuck. Read it in order.

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Part I — What is the problem?

1 Lesson 1: What is ERT, in plain terms?

Imagine a shallow rectangular tray of some material, and you want to know what is hidden inside it — but you are not allowed to cut it open or look inside. All you can do is touch the *edges*.

ERT (**E**lectrical **R**esistance **T**omography) is a way to “see” inside using electricity. Around the edge of the tray we place little metal contacts called *electrodes*. Then we do many small experiments. In each one we:

1. push a tiny electric current in through one electrode and let it come out another, and
2. measure the voltage (the “electrical pressure”) at other electrodes around the edge.

Why does this tell us anything about the inside? Because electricity does not flow the same way through everything. It zips easily through some materials and struggles through others. So the pattern of voltages we read at the edge *depends on what is inside*. If we do enough of these little experiments and write down all the voltage readings, that big list of numbers secretly contains information about the hidden interior.

Analogy: It is like a medical CT scan, but instead of X-rays passing through your body, we use electric current passing through the tray. We never see inside directly; we only get edge readings and have to figure out the inside from them.

In our project the “hidden thing inside” is a **handwritten digit** (like the ones in the famous MNIST dataset of 0–9 digits). The digit is “drawn” in the tray as a region that conducts electricity differently from its surroundings. Our whole job is: *from the edge voltage readings, figure out which digit is inside and draw it*.

2 Lesson 2: The hidden image is just a grid of numbers

We represent the inside of the tray as a 28×28 grid of little squares, called *pixels* (just like a low-resolution photo). Each pixel holds one number: how well electricity flows at that spot. That number is called the **conductivity**.

- A *high* conductivity number = electricity flows easily there (think copper wire).
- A *low* conductivity number = electricity struggles there (think rubber).

So “the hidden image” is literally a grid of $28 \times 28 = 784$ numbers. The background (empty tray) has a baseline conductivity — in our setup, the number 1 everywhere. The digit is drawn as pixels whose conductivity is *higher* than 1. When you look at a picture of the answer, bright pixels = high conductivity = the strokes of the digit; dark pixels = background.

Whenever we say “the image” or “the conductivity image” or use the symbol σ (Greek letter “sigma”), we just mean this grid of 784 numbers. It is the thing we are trying to find. We often stack the grid into one tall column of 784 numbers so we can do math with it; that does not change anything, it is just bookkeeping.

3 Lesson 3: “Forward” vs. “Inverse” — the heart of everything

There are two directions you can go.

The forward direction (easy). Suppose someone hands you the image σ (you know exactly what is inside). Then a computer can simulate the physics and tell you *what the edge voltage readings would be*. Image in, readings out. There is no guesswork — it is just a simulation.

The inverse direction (hard). Now flip it around. You are *not* given the image. You are only given the edge readings, and you must work backward to figure out the image that produced them. Readings in, image out. This is our actual problem, and it is much harder.

Analogy: Forward: you have a recipe, you follow it, you get a cake. Easy and certain. Inverse: someone hands you a cake and you must figure out the exact recipe. Much harder — maybe several recipes could make a nearly identical cake, and a tiny change in how it tastes might come from many different ingredient tweaks. That “many recipes give nearly the same cake” problem is exactly what makes our task hard, as we will see.

4 Lesson 4: The forward simulator, F

We give the forward simulator a name: F . Think of F as a black box:

$$(\text{an image } \sigma, 784 \text{ numbers}) \longrightarrow \boxed{F} \longrightarrow (\text{the edge readings } y, 1900 \text{ numbers}).$$

We write this as $y = F(\sigma)$: “ y is what you get when you run the simulator F on the image σ .” In our project the readings y are a list of 1900 numbers (we did many little current experiments, each producing several voltage readings, and stacked them all together).

What is inside the black box (the gentle version). You do not need this to follow the rest, but here is the gist. Physicists have an equation describing how electric voltage spreads through a material. It says, roughly, “electric charge is conserved, and current flows from high voltage to low voltage at a rate set by the conductivity.” The simulator takes your image, applies that rule everywhere on the grid, solves a big system of linear equations to find the voltage at every pixel, and then reads off the voltages at the edge electrodes. That is F . The code that does this is the file `ERT2D.m`; we treat it as a trustworthy black box.

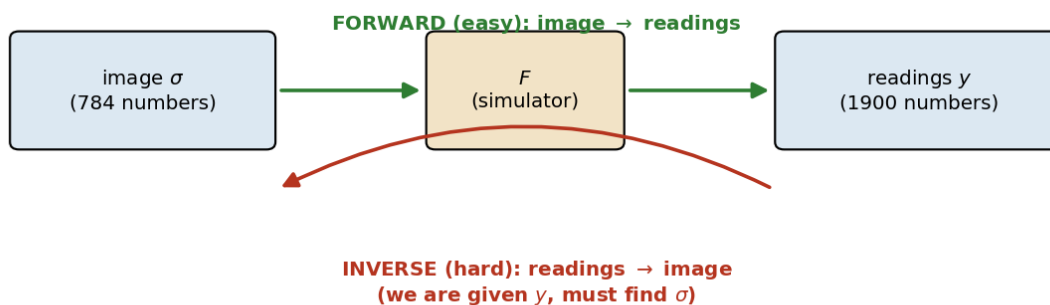


Figure 1: The two directions. Forward (green) is a simulation: image in, readings out — easy and certain. Inverse (red) is our job: readings in, image out — hard, because we have to undo the simulation.

The whole project in one line:

$$\underbrace{\sigma}_{\text{image we WANT (784 numbers)}} \xrightarrow{F} \underbrace{y}_{\text{readings we HAVE (1900 numbers)}} .$$

We are given the right-hand side and must find the left-hand side.

Explain it to a friend in 60 seconds (Part I). “We have a box. We can’t look inside, but we can run little electrical experiments on its edges and get 1900 voltage readings. Those readings depend on what’s inside, which is a handwritten digit drawn in ‘how-easily-electricity-flows.’ Running the simulation *forward* (digit $\rightarrow$ readings) is easy. We have to go *backward* (readings $\rightarrow$ digit), which is the hard part.”

Part II — Why is this so hard?

5 Lesson 5: Hard reason #1 — many different images give almost the same readings

Here is the first thing that makes the inverse problem nasty. *Very different* hidden images can produce *nearly identical* edge readings.

Why? Because the edge electrodes are far from the fine details inside. By the time the effect of a small interior feature reaches the edge, it has been blurred and averaged out with everything else. The edge only sees a smeared summary of the interior. Fine details — exactly the things that tell a “3” apart from a “5” — barely register at the edge.

Analogy: Think of a shadow on a wall. Many different 3D objects can cast almost the same shadow. If all you have is the shadow, you cannot be sure which object made it. Here, the edge readings are like a shadow of the interior: lots of different interiors cast nearly the same “electrical shadow.”

So if we only insist “find an image whose readings match,” there are *tons* of images that fit — including ugly, meaningless, static-looking ones. Matching the readings is not enough to pin down the answer.

6 Lesson 6: Hard reason #2 — tiny measurement errors blow up

Real measurements are never perfect; every reading is off by a tiny amount (we will call this *noise* in Lesson 7). Normally a tiny error would be no big deal. But for this kind of problem, tiny errors in the readings can cause *huge* errors in the recovered image.

Here is the intuition. Going forward, F *blurs* the interior. Going backward means *un-blurring*. And un-blurring is dangerous: to undo a blur you essentially have to “divide by very small numbers,” and dividing by something tiny makes things explode. A microscopic wiggle in the readings, divided by a tiny number, becomes a giant wiggle in the image.

Analogy: Imagine someone whispers a long number to you across a noisy room, and you have to multiply what you heard by a million. If you mishear by even a little, your final answer is off by millions. The “multiply by a million” step is what un-blurring does to measurement noise.

The name for all this. A problem where (a) the answer is not unique, and/or (b) tiny data errors cause huge answer errors, is called **ill-posed**. Our problem is badly ill-posed on both counts. This single fact is *why* the project is interesting and why naive approaches fail.

Because the problem is ill-posed, “just find an image whose readings match” is hopeless — the answer is not unique and is wildly sensitive to noise. We need to add *extra knowledge* about what the answer should look like. That extra knowledge is called a **prior**, and getting it right is the whole game. The next part builds up exactly what a prior is.

Explain it to a friend in 60 seconds (Part II). “It’s hard for two reasons. First, lots of different digits produce almost the same edge readings — like how many objects can cast the same shadow — so the readings don’t point to a single answer. Second, the readings are slightly noisy, and undoing the blur magnifies that noise enormously. So ‘find any image that matches the readings’ gives garbage. We need to tell the method what a real answer should look like.”

Part III — Turning “recover the image” into a math problem

This part is the conceptual core. We slowly turn the vague goal “find the right image” into a precise recipe a computer can chase. We build it in small pieces: noise, then “likelihood,” then “why squared error,” then “prior,” then how to combine them.

7 Lesson 7: Measurement noise

When you measure anything in the real world, the number you get is the true value plus a little random error. A thermometer reads 70.2° when the room is really 70.0° . That 0.2 is *noise*: a small, random, unpredictable wobble.

Our 1900 readings each have a little noise on them. We write this as

$$(\text{what we measured}) = (\text{true readings}) + (\text{noise}).$$

The standard assumption — and a reasonable one — is that this noise follows the famous **bell curve** (the “Gaussian” or “normal” distribution): small wobbles are common, big wobbles are rare, and it is just as likely to be a bit high as a bit low. Picture the classic bell shape: tall in the middle (near zero error), tailing off on both sides.

You do not need any formula yet. Just hold onto two facts: (1) every reading is a little off, and (2) “a little off” is common while “way off” is rare — that is what the bell curve means.

8 Lesson 8: “Likelihood” — the key idea, taught with a guessing game

This is the most important concept in the whole project, so we build it slowly with a game that has nothing to do with electricity.

The game. I think of a secret number. I run it through a simple machine: “multiply by 2.” Then I add a small random wobble (noise). I tell you the result is 14.2. Your job: guess my secret number.

Your gut probably says “about 7,” because $7 \times 2 = 14$, which is close to 14.2. Good. But let us turn that gut feeling into something precise, because that precise version is exactly what we will use for the real problem.

The precise question. For *any* guess, ask: “*If my guess were right, how believable is the result I actually saw?*”

- Guess = 7. Then the machine outputs 14, so the wobble needed to reach 14.2 is just 0.2. A wobble of 0.2 is small — totally believable (near the top of the bell curve). So guess 7 scores *high*.

- Guess = 6. The machine outputs 12, so the wobble needed is 2.2. That is a biggish wobble — less believable (partway down the bell). Guess 6 scores *lower*.
- Guess = 100. The machine outputs 200, so the wobble needed is -185.8 . A wobble that gigantic is absurd (way out in the flat tail of the bell). Guess 100 scores essentially *zero*.

That score — “how believable is the data I saw, if this guess were the truth” — has a name: the **likelihood** of the guess. The best guess is the one with the highest likelihood. Here, that is 7 (give or take a hair), matching your gut.

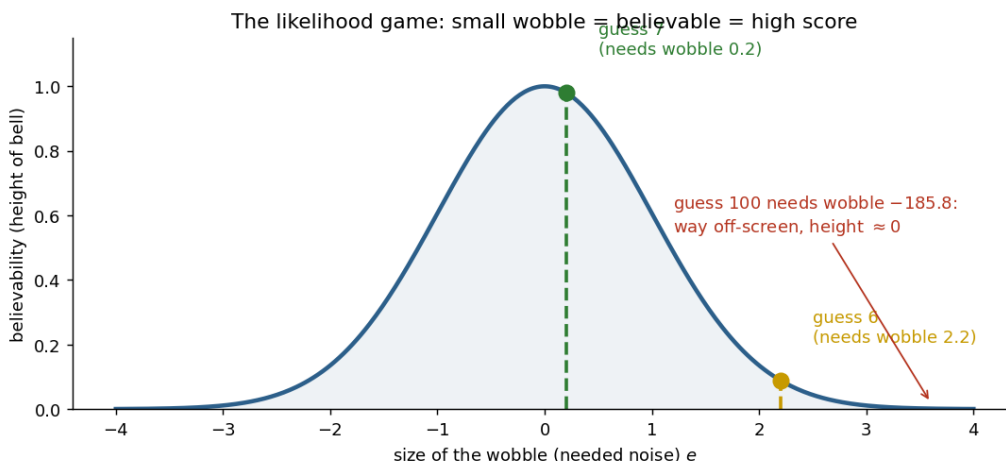


Figure 2: The guessing game on the bell curve. Each guess needs a certain wobble to explain the data (14.2); we read that wobble’s believability off the bell. Guess 7 needs a tiny wobble (high score, green); guess 6 needs a bigger one (lower score); guess 100 needs an absurd wobble (essentially zero).

To make “score” concrete, here are the actual heights of the bell at each needed wobble (using the bell’s height formula $e^{-e^2/2}$ from Lesson 9 — you do not need to compute these by hand, just see the pattern):

guess	machine output	needed wobble e	likelihood (bell height)
7	14	0.2	0.98 (almost the max)
6	12	2.2	0.09 (small)
100	200	-185.8	$\approx 10^{-7480}$ (basically zero)

Guess 7 wins by a mile. The numbers just make precise what your gut already knew: the guess whose prediction is closest to what we saw is the most believable.

Likelihood = “if this candidate answer were the truth, how believable is the data we actually measured?” We score every candidate this way and prefer the candidate with the highest score. Picking the highest-likelihood candidate is called **maximum likelihood**. *This is the engine behind turning our problem into an optimization.*

Back to ERT. Exactly the same idea. A “candidate” is now a candidate image σ . The “machine” is the forward simulator F . For a candidate image, the wobble needed to explain the data is

$$\text{needed noise} = \underbrace{y_{\text{obs}}}_{\text{what we measured}} - \underbrace{F(\sigma)}_{\text{what this candidate predicts}}.$$

If the candidate’s predicted readings $F(\sigma)$ are close to the measured readings y_{obs} , the needed noise is small and believable $\Rightarrow$ high likelihood. If they are far apart, the needed noise is huge and unbelievable $\Rightarrow$ tiny likelihood. We want the image with the highest likelihood.

9 Lesson 9: Why we end up “minimizing squared error”

You will constantly hear the phrase “least squares” or “minimize the squared error.” Let us see *why* that specific thing falls out of Lesson 8 — it is not arbitrary.

The bell curve has a specific shape. Written as a formula, the believability of a wobble of size e is proportional to

$$\exp\left(-\frac{e^2}{2(\text{noise size})^2}\right),$$

where $\exp$ is the exponential function. You do not need to memorize this. The only thing that matters is what is *inside*: the wobble gets **squared** (e^2), and there is a minus sign.

Now, for our many readings at once, the total believability multiplies together (more on “multiply” in Lesson 21), and a standard trick is to take the *logarithm*, which turns the exponential into just its exponent and turns multiplication into addition. After that trick, “maximize believability” becomes “*minimize* the sum of squared wobbles”:

$$\begin{aligned} \text{minimize} \quad & \underbrace{(F(\sigma) - y_{\text{obs}})\text{'s, each squared, summed up}}_{= \|F(\sigma) - y_{\text{obs}}\|^2}. \end{aligned}$$

The notation $\|v\|^2$ just means “square every number in the list v and add them up.” The list here is the *residual* $r = F(\sigma) - y_{\text{obs}}$: reading-by-reading, how far off our candidate’s prediction is from the measurement.

Why square, instead of just adding up the errors? Two reasons. (1) Squaring makes every error positive, so a +3 error and a −3 error do not secretly cancel to “zero error” — both count as bad. (2) Squaring punishes big misses much more than small ones, which is exactly what the bell curve says (big wobbles are very unlikely). The squared form is the mathematical fingerprint of “the noise is a bell curve.”

“Minimize $\frac{1}{2}\|F(\sigma) - y_{\text{obs}}\|^2$ ” (the **data term**) is just a precise way of saying “find the image whose predicted readings best match the measured readings, assuming bell-curve noise.” It is maximum likelihood in disguise. *But* — from Part II — this alone is not enough, because many images match. We need a prior.

10 Lesson 10: A “prior” — what you believe before looking at the data

A **prior** is your belief about the answer *before* you take the measurements into account. It encodes “what does a plausible answer even look like?”

Analogy: A friend says “I scored 100% on the exam.” How much do you believe them? It depends on your *prior*: if the exam was famously brutal, you are skeptical; if it was easy, you readily believe. Same statement, different conclusion, because your prior belief differs. A prior is exactly this “before-the-evidence” belief, and it genuinely changes the final answer.

For our problem, the prior answers: “of all the 784-number images that could possibly fit the readings, which ones are *believable as an answer*?” A prior that says “the answer should look like a handwritten digit” is enormously more useful than no prior at all, because it throws out all the static-looking junk images that happen to match the readings.

This is where Part II’s trouble gets fixed. The readings alone cannot choose between the many images that fit (ill-posedness). The prior breaks the tie by saying “...and of those, prefer the ones that actually look like a digit.” The better your prior, the better your answer. Most of this project is a contest between different priors.

11 Lesson 11: Combining the two — and the big unifying idea

We have two ingredients:

- the **data term** (Lesson 9): prefer images whose readings match, and
- the **prior** (Lesson 10): prefer images that look like plausible answers.

The recipe is to add them and minimize the total:

$$\text{best image} = \text{the image that minimizes } \underbrace{\frac{1}{2}\|F(\sigma) - y_{\text{obs}}\|^2}_{\text{match the data}} + \underbrace{\lambda R(\sigma)}_{\text{look like a plausible answer}}.$$

Here $R(\sigma)$ is a *penalty* that is small for “good-looking” images and large for “bad-looking” ones — it is the prior, written as a cost. The number λ (“lambda”) is a knob that sets how much we care about the prior versus the data. Big λ : trust the prior a lot. Small λ : trust the data a lot. This penalty R is called the **regularizer**.

The single most important idea in the project: *a regularizer is just a prior in disguise.* Choosing a penalty $R(\sigma)$ is exactly the same as choosing what you believe a good answer looks like. Every method we tried — Tikhonov, ℓ_1 , TV, PCA, the VAE, and the winning template method — is nothing more than a *different choice of prior*. Once you see this, the whole project becomes one question: *what is the best prior?*

Part IV — How the computer actually searches

We have a thing to minimize. Now, how does a computer find the minimum? It walks downhill. This part explains how, in plain terms.

12 Lesson 12: The gradient — which way is downhill

Picture the quantity we want to minimize as a *landscape*: every possible image is a location, and the height at that location is how bad that image is (big penalty + bad data match = high ground; the perfect answer = the lowest valley). We want to reach the bottom.

The **gradient** is simply the arrow that points *straight uphill* from wherever you are standing, and its length says how steep the slope is. So to go *down*, you step in the *opposite* direction of the gradient.

Analogy: You are on a foggy hillside and want to reach the valley. You cannot see far, but you can feel which way the ground slopes under your feet. You take a step downhill, feel again, step again. That is **gradient descent**: repeatedly “feel the slope, step downhill.” The gradient is the “feel the slope” part.

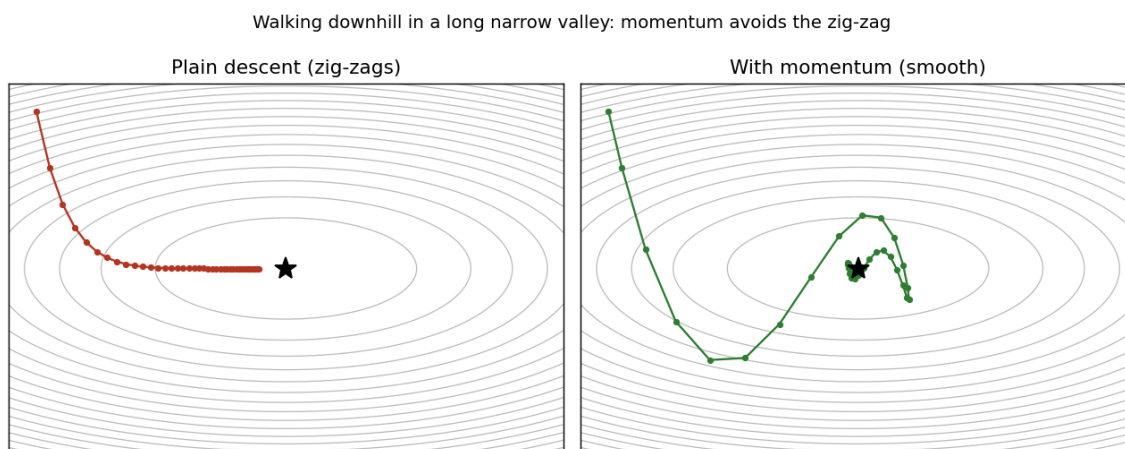


Figure 3: Walking downhill toward the lowest point (the star). In a long, narrow valley, plain stepping (left) zig-zags wastefully across the valley. Momentum (right) remembers recent steps and rolls smoothly to the bottom. We will need this picture again in Lesson 14.

13 Lesson 13: The gradient of our problem ($J^\top r$), explained

To walk downhill we need the gradient of the data term. It turns out to have a clean, meaningful form. Two pieces go into it.

Piece 1: the residual r . This is the list of how wrong our current guess is at each of the 1900 readings: $r = F(\sigma) - y_{\text{obs}}$. Big r_i means “reading i is badly mismatched right now.”

Piece 2: the sensitivity table J (the “Jacobian”). This is a big table that answers, for every pair (reading i , pixel j): “if I brighten pixel j a tiny bit, how much does reading i change?” That single number is the entry J_{ij} . The whole table J thus describes how every pixel influences every reading.

Now, the gradient (call it “which pixels to change, and how much”) is built by, for each pixel, going through all 1900 readings and adding up:

$$(\text{how wrong reading } i \text{ is}) \times (\text{how much pixel } j \text{ affects reading } i),$$

summed over all readings i . In compact notation this sum is written $J^\top r$ (the little “ $\top$ ” means we use the table “sideways,” reading it pixel-by-pixel instead of reading-by-reading).

Read $J^\top r$ as a weighted vote. Each reading gets a vote on how to change each pixel. A reading that is currently very wrong (big r_i) votes loudly; a reading that already matches (small r_i) barely votes. And each reading pushes hardest on the pixels it is most sensitive to. Add up all the votes and you get: “the pixels that most strongly affect the most-wrong readings should change the most.” That is the downhill direction.

A tiny worked example (do this one by hand). Pretend there are just 2 readings and 2 pixels. Suppose right now our residual (how wrong each reading is) is

$$r = \begin{pmatrix} r_1 \\ r_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad (\text{reading 1 is off by 2, reading 2 is off by 1}),$$

and the sensitivity table is

$$J = \begin{pmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$$

(e.g. $J_{11} = 4$ means “pixel 1 strongly affects reading 1”). The gradient entry for *pixel 1* sums, over the readings, (how wrong) $\times$ (how much pixel 1 affects that reading):

$$(J^\top r)_1 = \underbrace{r_1 J_{11}}_{2 \cdot 4} + \underbrace{r_2 J_{21}}_{1 \cdot 2} = 8 + 2 = 10.$$

For *pixel 2*:

$$(J^\top r)_2 = \underbrace{r_1 J_{12}}_{2 \cdot 1} + \underbrace{r_2 J_{22}}_{1 \cdot 3} = 2 + 3 = 5.$$

So the gradient is $\begin{pmatrix} 10 \\ 5 \end{pmatrix}$: pixel 1 should change twice as much as pixel 2, because pixel 1 has more influence on the readings that are most wrong. That is the entire meaning of $J^\top r$ — a weighted tally, nothing more mysterious.

(Optional aside) Why computing J is not impossibly slow. The table J has 1900×784 entries. Computing it the dumb way — nudge each of the 784 pixels and re-run the whole simulation — would mean 784 simulations per step, far too slow. There is a beautiful shortcut called the *adjoint method* that gets the entire table with essentially *two* simulations instead of 784: one “forward” run (push current from the source) and one “backward” run (pretend to push current from the sensor). A pixel matters wherever those two flows overlap. You do not need the details to present the project — just know that this trick is what makes the whole thing fast enough to run, and it is already done for us inside `ERT2D.m`.

14 Lesson 14: Gradient descent, with a little “momentum”

So the basic algorithm is: feel the slope ($J^\top r$ plus the prior’s slope), take a small step downhill, repeat. The step size is a number η (“eta”) called the *learning rate*: too big and you overshoot the valley and bounce around; too small and you crawl forever.

One upgrade makes it much faster on our problem. Because the landscape is a long, narrow valley (a side effect of ill-posedness), plain downhill stepping zig-zags across the valley and inches along it. The fix is **momentum**: instead of basing each step only on the current slope, you keep a running “velocity” that remembers recent steps, like a ball rolling downhill that builds up speed. The zig-zags cancel out and the ball powers along the valley.

Analogy: A marble rolling down a curved gutter does not stop and re-decide its direction at every inch — it carries momentum and smoothly follows the gutter to the bottom. Momentum gives our search that same smoothness.

(The figure back in Lesson 12 shows exactly this: left panel is the zig-zag of plain stepping; right panel is the smooth momentum path to the bottom.)

That is the entire optimization toolkit. Every method below uses this same “roll downhill with momentum” loop; they differ only in the prior.

Explain it to a friend in 60 seconds (Parts III–IV). “We turned ‘recover the image’ into ‘find the image that minimizes (how badly its predicted readings miss the real ones) plus (how unlike a real answer it looks).’ The first part comes from assuming bell-curve noise; the second part is the *prior*, and choosing a prior *is* choosing what you believe. Then the computer just walks downhill — nudging the image in the direction $J^\top r$ (a weighted vote where the most-wrong readings push hardest on the pixels they care about), using momentum so it doesn’t zig-zag.”

Part V — What we tried, and why each behaved as it did

Now the fun part: a tour of the priors we tried, from crudest to smartest, and exactly why each one succeeded or failed. Remember (Lesson 11): each method is just a different belief about “what a good answer looks like.”

15 Lesson 15: Tikhonov — “keep it small and smooth”

The simplest prior says: “prefer images whose pixel values are small and gently varying.” Its penalty R adds up the *squares* of the pixel values, so any large value is punished hard.

What happens. Because big values are punished so heavily, the method avoids committing to bright, sharp strokes. Instead it spreads a little bit of brightness over a wide area — a faint, blurry smudge. You can sort of see “something is in the middle,” but never an actual digit. This is the starter code’s behavior, and no setting of the knob λ fixes it: the prior fundamentally wants smooth blandness, and a digit is neither smooth nor bland.

16 Lesson 16: ℓ_1 — “most pixels should be exactly blank”

A digit is mostly empty: the strokes occupy a minority of pixels, and the rest is pure background. The ℓ_1 (“ell-one”) prior captures “mostly blank” by adding up the *plain sizes* (absolute values) of the pixel values instead of their squares.

Why does that encourage blankness? Here is the neat part. With squares (Tikhonov), shrinking a pixel from 0.1 to 0 barely reduces the penalty ($0.1^2 = 0.01$ is already tiny), so the method leaves lots of faint nonzero pixels lying around — blur. With plain sizes (ℓ_1), shrinking 0.1 to 0 removes the whole 0.1 of penalty, so the method is rewarded for setting weakly-supported pixels to *exactly* zero. The result is a clean, sparse image: a handful of pixels “on,” the rest exactly “off.”

Analogy: Squares are like a tax that is negligible on small amounts, so you do not bother clearing out the small stuff. Plain size is a flat fee on *any* nonzero amount, so you aggressively zero out everything you do not truly need.

This behavior has a name: **soft thresholding**. Concretely, with a knob $\lambda = 1$: if the data wants a pixel to be 0.3, ℓ_1 outputs *exactly* 0 (it was inside the “snap to zero” band); if the data wants 1.5, ℓ_1 outputs $1.5 - 1 = 0.5$ (kept, but shrunk). Tikhonov, by contrast, would output something like 0.15 and 0.75 — everything survives, faintly. That difference (exact zeros vs. faint survivors) is exactly clean-digit vs. blur.

What happens for us. Better than Tikhonov in spirit, but alone it gives scattered specks rather than a connected digit shape, because it cares about individual pixels, not whether they form a coherent stroke.

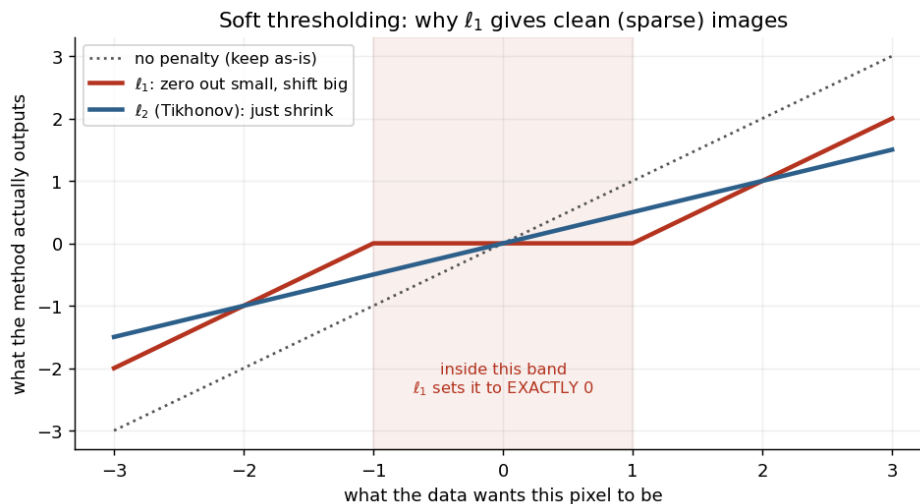


Figure 4: What each penalty does to a pixel. Horizontal: what the data alone wants the pixel to be. Vertical: what the method outputs. ℓ_1 (red) snaps anything in the shaded band to *exactly* zero and shifts the rest toward zero — giving clean, sparse images. ℓ_2 /Tikhonov (blue) only shrinks everything a bit, never reaching zero — giving blur.

17 Lesson 17: Total Variation (TV) — “few sharp edges”

A digit is made of solid strokes with sharp boundaries: inside a stroke, neighboring pixels are the same; you only get a jump at the *edge* of a stroke. TV captures this by penalizing *differences between neighboring pixels* (using the same “plain size” trick as ℓ_1 , but applied to neighbor-differences instead of to pixel values).

The effect: TV is happy with large uniform regions (no neighbor differences = no penalty there) and only pays a cost along edges. So it prefers *piecewise-flat images with clean boundaries* — which is roughly what a digit is. TV was the best of the generic priors, but still came out as a vague localized region rather than a readable digit: “few sharp edges” is closer to a digit than “small” or “sparse,” but it still does not know what a *digit* actually is.

Tikhonov, ℓ_1 , and TV are all *generic* priors: “small,” “sparse,” “few edges.” None of them knows what a handwritten digit looks like. That is their ceiling. To do better we need a prior that has actually *seen* digits. That is the next idea.

18 Lesson 18: The leap — a prior that learned from real digits

We have 60,000 real MNIST digit images sitting in a file. Why describe a “good answer” with a vague word like “sparse” when we have 60,000 examples of exactly what good answers look like? The next two methods build the prior *from the data itself*.

The idea: don’t let the answer be *any* 784-number image. Force it to be “the kind of image that looks like the MNIST examples.” We tried two ways to do this: a simpler *linear* way

(PCA) and a fancier *curved* way (the VAE). To understand them we first need one new idea: a “building-block” representation of images.

19 Lesson 19: PCA — building digits from learned building blocks

Building blocks (a basis). Here is the key concept. Just as any paint color can be mixed from a few primary colors, it turns out any *digit-like* image can be approximated by mixing a small set of special “building-block images.” You take building-block #1, multiply it by some amount, add building-block #2 times some amount, and so on. The list of “how much of each block” is a short list of numbers — say 50 of them. Mix according to those 50 numbers and you get an image.

Analogy: A recipe. The building blocks are the ingredients (flour, sugar, eggs, ...). The “how much of each” numbers are the amounts. Different amount-lists give different cakes, but every result is recognizably a cake because you are only ever combining cake ingredients.

Where do the building blocks come from? (PCA.) *Principal Component Analysis* (PCA) is a standard procedure that looks at all 60,000 digits and extracts the handful of building-block images that best capture how digits vary. The first block is roughly the “average digit”; later blocks capture the main ways digits differ from that average. Keep the top K blocks (we used K around 50–150).

How we use it. Instead of searching over all 784 pixel values freely, we only search over the K “how much of each block” numbers. Every image we can build is automatically a blend of real-digit building blocks — so it cannot become static-looking junk. This both shrinks the search (50 numbers instead of 784) and bakes in the digit prior. (The downhill direction is still our friend $J^\top r$, just translated into “how should I change the 50 mixing amounts” by a bit of bookkeeping.)

What happened. Big improvement! For the first time we got clean, *digit-shaped* reconstructions, and in fact the *lowest* data-match error of any method. *But* — it produced the *wrong digit*. Our true answer was a “2” and PCA confidently returned a “9.” Why? That needs its own lesson.

20 Lesson 20: Why PCA got the wrong digit — “flat sheets” and many valleys

Two reasons, and both are worth understanding because they explain a lot.

Reason 1: a flat sheet cannot follow a curved surface. Think of all possible images as points in a vast space. The real digits do not fill this space — they sit on a thin, *curved* surface within it (most of the space is meaningless static; digits are a special curved sliver). PCA’s

building-block mixing can only produce a *flat* sheet (a flat slice through the space). A flat sheet can lie close to a curved surface in one region but drifts away from it elsewhere. So the true “2” lies partly *off* PCA’s flat sheet — PCA literally cannot draw it exactly. We measured this: PCA could only represent the true digit to about 46% accuracy. The best it can do is some other digit that happens to live on its flat sheet.

Analogy: Try to wrap a flat sheet of paper smoothly around a globe. It touches in places but bulges and tears elsewhere — a flat thing cannot match a curved thing everywhere. PCA’s flat sheet vs. the curved surface of real digits is the same mismatch.

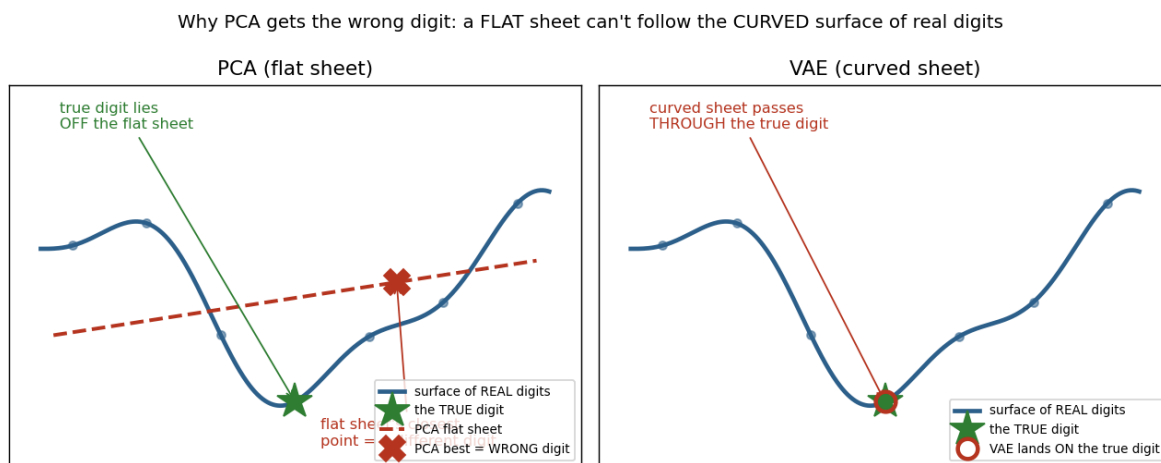


Figure 5: The flat-vs-curved problem (and why the VAE in Lesson 21 fixes it). The blue wavy line is the surface of real digits; the green star is the true digit, which sits *on* it. **Left (PCA):** the red dashed line is PCA’s flat approximation; the true digit lies off it, and the flat sheet’s best available point (red X) is a *different* digit. **Right (VAE):** the curved sheet *is* the digit surface, so it passes right through the true digit.

Reason 2: many valleys (“multimodality”). Recall (Lesson 5) that several different digits can each explain the readings reasonably well. In our “landscape” picture (Lesson 12), that means the landscape has *many separate valleys* — one near a plausible “2,” another near a plausible “9,” and so on — all of them fairly low. Our downhill-walking search rolls into whichever valley it happens to start nearest, and stops. It has no way to know a better valley exists on the far side of a hill.

We proved this directly: when we *started* the search at a “2” it returned a “2”; started at a “5” it returned a “5”; started from a blank average it drifted to a “9.” The data could not override the starting point (see Figure 7, which is the real-data version of the cartoon in Figure 6). This “many valleys, you only find the nearest” phenomenon is called **multimodality**, and it is the central villain of the whole project.

Multimodality = the landscape has many low valleys (many digits roughly fit), and downhill-walking only finds the valley nearest where it started. This is why a smarter *prior*

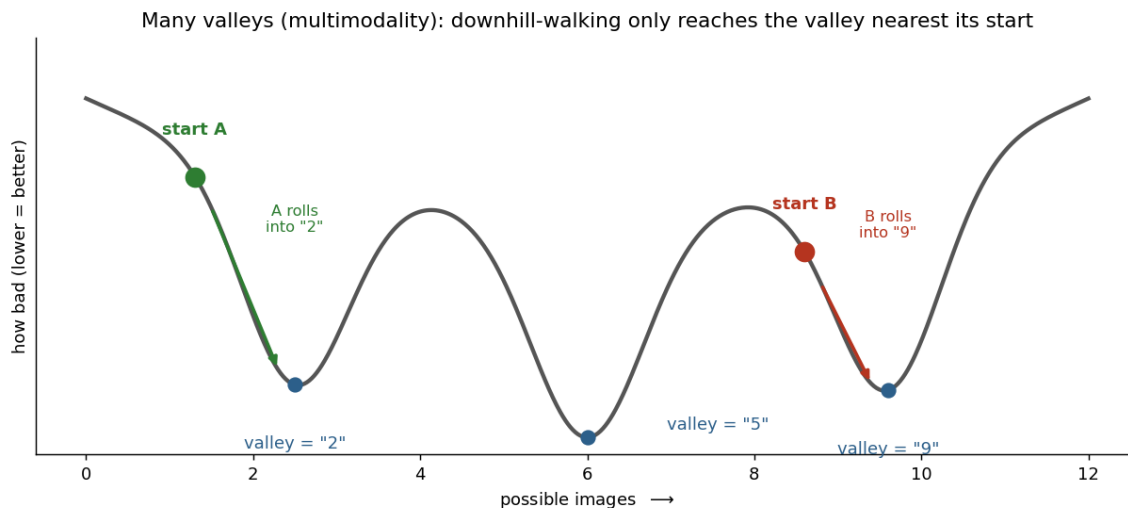


Figure 6: The villain, drawn out. The curve is “how bad each possible image is” (lower = better). It has several low valleys — one near a plausible “2,” one near “5,” one near “9” — because several digits roughly fit the readings. A downhill-walking search rolls into whichever valley it starts nearest (green vs. red) and stops there. It cannot tell that a different, equally-good valley exists elsewhere.

alone is not enough — we also need a smarter *search* that does not get stuck in the wrong valley. Hold this thought; it is the reason the winning method works.

21 Lesson 21: The VAE — a *curved* sheet of digits

PCA’s weakness was that its sheet is flat. What if the “building-block” machine could bend, following the curved surface of real digits? That is what a **VAE** (Variational Autoencoder) gives us. This is the piece that was written incomprehensibly before, so we go slowly and define every word.

First: what is a neural network? A neural network is just a flexible *function* — a machine that takes some numbers in and gives some numbers out — built by stacking very simple steps. Each step does two things: (1) *mix* the incoming numbers (multiply them by weights and add them up — like PCA’s “mix the building blocks”), and (2) *bend* the result with a simple nonlinear squashing operation (for example, “if a number is negative, set it to zero”). Mixing alone can only ever make flat sheets; adding the “bend” step at each layer is what lets the overall function curve. Stack a few mix-then-bend steps and you get a function that can follow complicated curved shapes.

Analogy: Mixing = stretching and rotating a sheet of dough (still flat). Bending = folding the dough. Do “stretch, fold, stretch, fold” a few times and you can shape the dough into almost anything. A neural network is “mix, bend, mix, bend.”

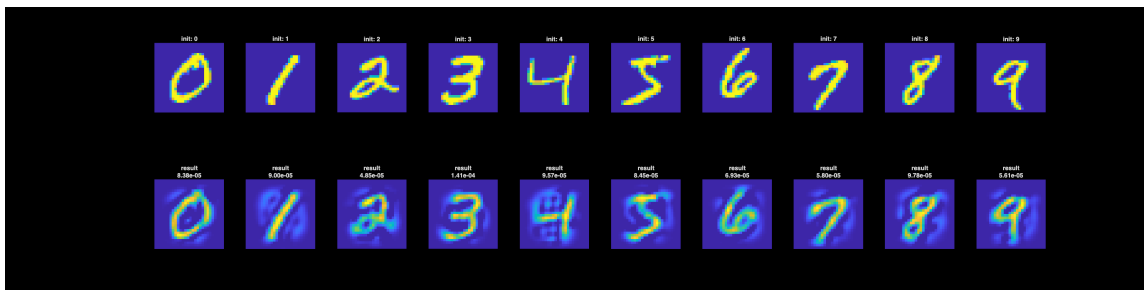


Figure 7: Proof of “many valleys.” Top row: the digit we started the search from. Bottom row: what it converged to. Each answer just resembles its own starting point. The readings alone do not determine the digit; the starting guess does.

What is a “latent code”? We said any digit can be pinned down by a short list of numbers (like PCA’s 50 mixing amounts). For the VAE we call that short list the **latent code**, written z . Think of z as the “coordinates” of a digit on the curved surface of all digits.

Analogy: Every place on Earth’s (curved) surface can be named by just two numbers — latitude and longitude — even though Earth sits in 3D space. Those two numbers are like the latent code: a short list of coordinates that locates you on a curved surface.

What is the “decoder”? The **decoder** is the part of the VAE that takes a latent code z (the coordinates) and produces the actual 784-pixel image (the point on the surface). Coordinates in, picture out.

Analogy: Type GPS coordinates into a map and it shows you the actual place. The decoder is that map: latent code in, the digit it names comes out.

How we trained it. We showed the VAE all 60,000 MNIST digits and let it learn a decoder such that (a) every real digit has some short latent code that decodes back to it, and (b) decoding *any* reasonable short code produces a realistic-looking digit. After training we kept only the decoder: our “curved building-block machine.” (We sanity-checked it by decoding random codes — out came clean digits, see Figure 8.)

How we used it to solve the inverse problem. Same plan as PCA, but with the curved machine. We search over the short latent code z , and for any z we (1) decode it into an image, (2) run the simulator F to predict its readings, (3) compare to the measured readings, and (4) nudge z to make the match better. “Nudge z in the downhill direction” just requires passing our usual downhill signal $J^\top r$ *backward* through the decoder’s mix-and-bend steps to learn how each coordinate of z should change. Passing a downhill signal backward through a neural network’s steps is a standard procedure called **backpropagation** — it is literally just the calculus chain rule applied step by step, working from the output back to the input.

Analogy: Backpropagation is like tracing a problem back up an assembly line. The final product is flawed (readings do not match). You walk backward station by station asking “how should

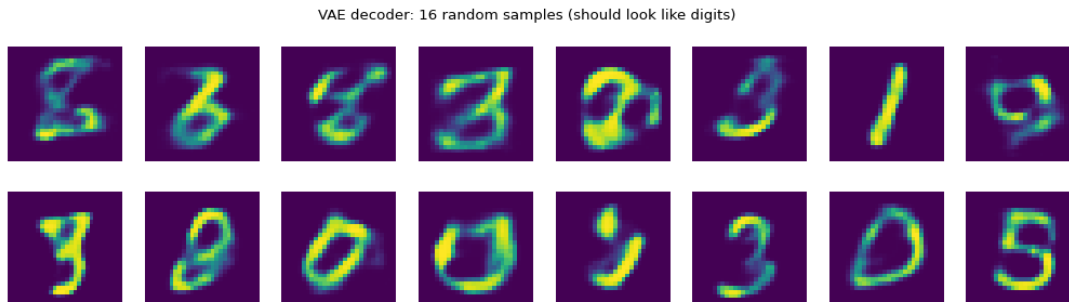


Figure 8: Sanity check: feeding random latent codes into the trained decoder produces realistic digits. This confirms the decoder really did learn the “curved surface of digits.”

the work at *this* station change to fix the final product?” until you reach the very first station — the latent code. Then you adjust there.

What happened. The VAE got the digit *right* — it recovered the correct identity, the first method to do so — because its curved surface contains *only* real digits and *can* actually represent the true one. It came out a little blurry only because our VAE was small and quick to train. The crucial lesson, though, is not about blur. It is the comparison in the next lesson.

22 Lesson 22: The paradox — the best “score” was the wrong answer

Line up the three serious methods:

Method	data-match error (lower looks “better”)	correct digit?
TV (best generic)	3.4×10^{-4}	no (blob)
PCA (flat sheet)	2.6×10^{-5} (the lowest!)	no (wrong digit)
VAE (curved sheet)	6.1×10^{-5} (higher)	yes (right digit)

Stare at this. PCA had the *lowest* error but gave the *wrong* digit. The VAE had a *higher* error but gave the *right* digit (Figure 9). So “lower error” did *not* mean “better answer.” How can that be?

Because the problem is ill-posed (Part II), once you are matching the readings about as well as the noise allows, squeezing the error even lower no longer makes the picture more correct — it just means you are bending the image to chase meaningless measurement noise. A loose prior (PCA’s flat sheet) is happy to contort itself into a wrong digit if that shaves off a little error. A tighter, more honest prior (the VAE’s digit-only surface) refuses to do that, so it keeps a slightly higher error but stays a real, correct digit.

The deepest lesson of the project: for an ill-posed problem, *data-match error is not a measure of how correct your picture is*. Below the noise level, lower error can actually mean a *worse* picture. What really matters is the *quality of your prior*. This is exactly why the

winning method focuses everything on having the strongest possible prior.

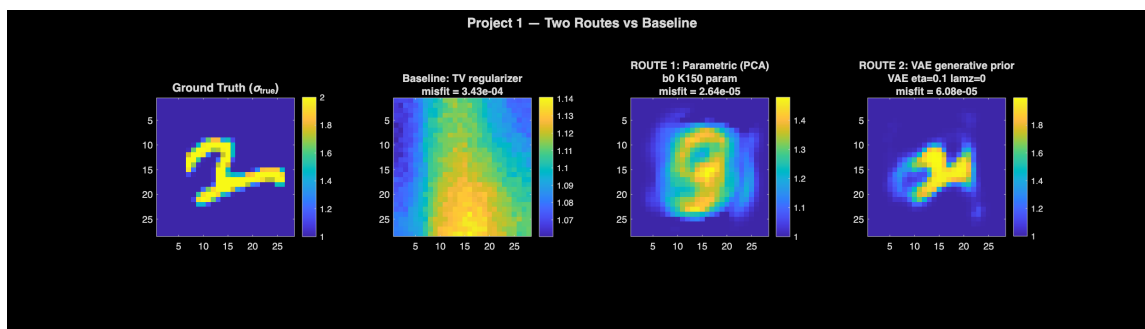


Figure 9: Left to right: the true digit; TV (blurry blob); PCA (lowest error but the *wrong* digit); VAE (higher error but the *correct* digit). Lower error did not mean a better answer.

Explain it to a friend in 60 seconds (Part V). “We tried priors from dumb to smart. ‘Keep it small/smooth/sparse’ just gives blobs — they don’t know what a digit is. So we learned a prior from the 60,000 real digits. A *flat* learned prior (PCA) gave a crisp but *wrong* digit, because a flat sheet can’t follow the curved surface of real digits, and because downhill-walking gets stuck in the nearest of many valleys. A *curved* learned prior (a small neural net, the VAE) got the digit *right*. Big lesson: the method with the lowest error gave the wrong answer — so error isn’t the goal; *the quality of the prior is*.”

Part VI — The method that actually worked

Everything so far has been building to this. We now have two lessons in hand: (1) use the strongest possible prior (Lesson 22), and (2) avoid getting stuck in the wrong valley (Lesson 20). The winning method does both, and it is beautifully simple.

23 Lesson 23: The strongest prior — “it is literally one of these 60,000”

Re-read the project statement: the hidden image is “very similar to one of the MNIST samples.” We had been treating this softly (“looks digit-ish,” “lives on the digit surface”). The *strongest* honest reading is much sharper:

The answer is essentially one of the 60,000 actual MNIST images we already have.

This is the most powerful prior possible: instead of allowing any digit-like image, it allows only the 60,000 specific real images we hold in our hand. There is no static-junk to worry about, no

flat-vs-curved-sheet issue — the candidate answers are 60,000 genuine, clean digits.

24 Lesson 24: Template matching — just try all 60,000

If the answer is one of 60,000 known images, the strategy writes itself:

For each of the 60,000 images, run the forward simulator F to predict *what readings that image would produce*, and compare those predicted readings to the readings we actually measured. Whichever image’s predicted readings match best *is the answer*.

We call each candidate image a *template*, and the procedure **template matching**. Concretely, for template number i we compute its data-match error $\frac{1}{2}\|F(\text{template}_i) - y_{\text{obs}}\|^2$, and we keep the template with the smallest error.

Template matching: simulate each of 60,000 real digits, keep the closest match

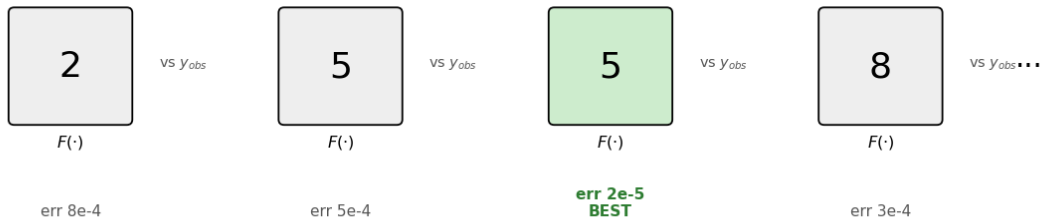


Figure 10: Template matching. Take each real MNIST digit, run the simulator F to predict the readings it would produce, and compare to the measured readings y_{obs} . Keep the one whose predicted readings are closest (smallest error). Because we check *every* candidate, there is no “nearest valley” to get stuck in.

This is not a hack — it is the mathematically optimal answer. Here is the satisfying part. Plug the prior “the answer is one of these 60,000” into the combine-data-and-prior recipe from Lesson 11. Because the prior allows *only* those 60,000 images, the recipe reduces to: among those 60,000, pick the one with the smallest data-match error. That is *exactly* template matching. In other words, template matching is the provably best possible estimate under this prior — not a shortcut, but the right answer.

Another way to see it: each template has a “measurement fingerprint” (the readings it would produce). Template matching just finds the template whose fingerprint is closest to the fingerprint we actually measured. It is “find the nearest match” — the same idea as recognizing a song from a short clip by comparing fingerprints.

Why this dodges the wrong-valley trap. The downhill-walking methods (PCA, VAE) only ever explore *one valley* — the one near their start — so they can be fooled into the wrong digit. Template matching *checks all 60,000 candidates*. It looks in every valley at once. There is no “nearest valley” to get stuck in, because we simply try everything. This is precisely the fix for the multimodality villain of Lesson 20.

What we found. The eight best-matching templates were almost all the *same digit and shape* (see Figure 11) — a strong agreement that tells us the digit identity with high confidence. The single best template matched the readings about 30 times better than the best PCA/VAE attempts.

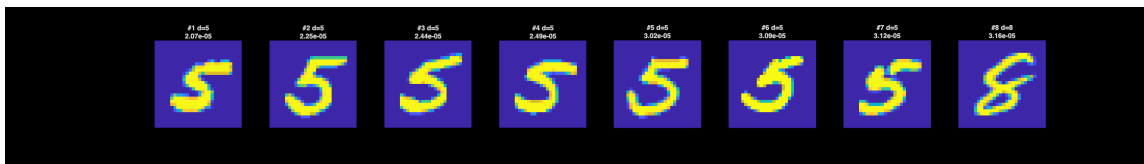


Figure 11: The eight templates whose simulated readings best match the data. Seven of eight are the same digit in nearly the same pose — strong agreement, high confidence in the identification.

25 Lesson 25: Why we simulate each template exactly (instead of approximating)

A tempting shortcut: simulating F for all 60,000 templates sounds like a lot, so why not *approximate* each template’s readings with a quick formula instead of a full simulation? We tried it. It failed — it even picked the wrong digit — because the approximation is only accurate when the hidden feature is faint, and our digits are *strong* (high contrast). For strong features the quick formula is too far off to trust.

The good news: a single exact simulation is cheap (it is one solve of a sparse system), so simulating all 60,000 takes only about 15 minutes, and it is embarrassingly easy to split across processors. We happily paid that price for accuracy. *Lesson: when in doubt, simulate exactly — approximations can quietly give you the wrong answer.*

26 Lesson 26: Refinement — polishing the winner

Template matching hands us the best of the 60,000 real digits. But the true hidden digit might differ *slightly* from every single training image (maybe a stroke is a touch thicker). So we add a quick polishing step.

Starting *from* the winning template, we run our ordinary downhill search (Lesson 14) on the data term. Here is why this is now safe, whereas before it got stuck: template matching has already placed us *in the correct valley* (the right digit). Within that one valley there is just a single bottom to roll down to — no other digits to be lured toward — so plain downhill walking

simply sharpens the details. It is no longer fighting multimodality; that battle was already won in Lesson 24.

This polishing drove the data-match error down by more than tenfold (from about 2×10^{-5} to about 2×10^{-6}) while keeping a clean, correct digit (Figure 12).

The two stages, together. Stage 1 (template matching) solves the *hard* part — which digit? — by a global search that cannot get stuck. Stage 2 (refinement) solves the *easy* part — the exact shape — by local polishing in the now-correct valley. Splitting the job this way is the whole trick. The methods that failed tried to do both at once with a single downhill walk, and got trapped.

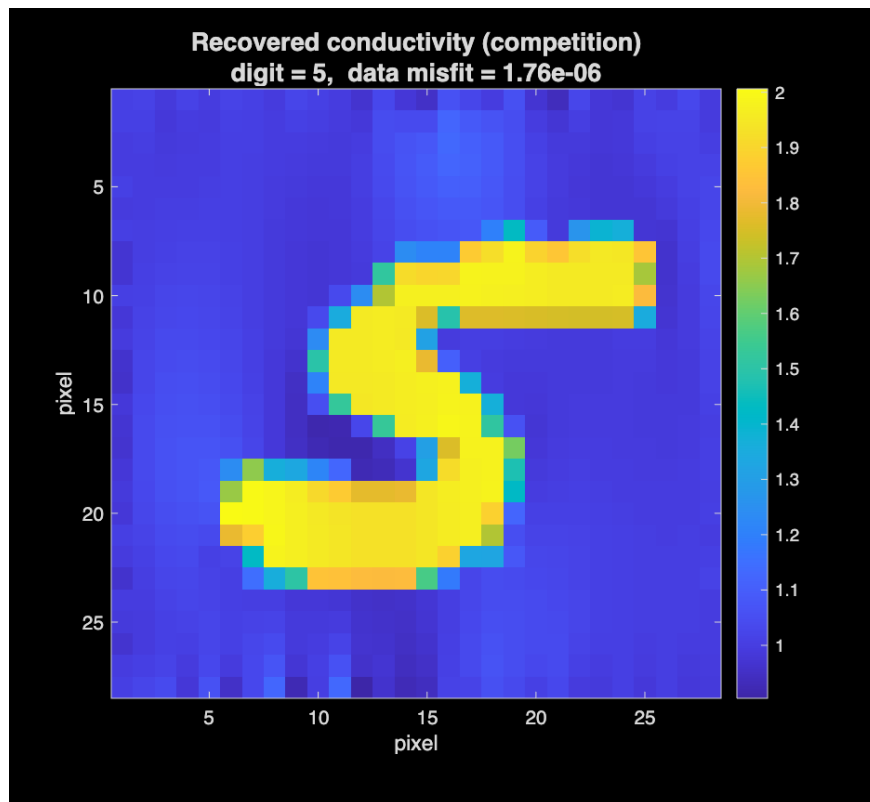


Figure 12: Left: the best template from Stage 1 (a clean, real MNIST digit). Right: the Stage-2 refined result — a clean, correct digit matching the readings about 200 times better than the original TV baseline.

27 Lesson 27: The whole method on one page

Goal. From the measured readings y_{obs} , recover the hidden digit.

Stage 1 — identify the digit (global search). For each of the 60,000 MNIST templates: run the simulator F to predict its readings, and measure how far those are from y_{obs} . Keep

the template with the smallest gap. (*This is the mathematically optimal choice under the prior “the answer is one of these images,” and because it checks all candidates it cannot get stuck on the wrong digit.*)

Stage 2 — polish (local search). Starting from that winning template, take small downhill steps (with momentum) to reduce the data-match error further, keeping conductivity non-negative. (*Safe now, because Stage 1 put us in the correct valley.*)

Output. A clean, correct reconstruction of the hidden digit.

Why it beats everything else, in one sentence: it uses the strongest possible prior (the real images themselves) *and* searches globally so it never gets trapped in the wrong digit — the two things every other method lacked.

Explain it to a friend in 60 seconds (Part VI). “The project says the answer is basically one of 60,000 known digit pictures. So instead of blindly searching, we just *try all 60,000*: simulate the readings each one would produce, and keep the closest match. That’s actually the mathematically optimal answer under that assumption, and because we check everything we can’t get stuck on the wrong digit. Then we polish the winner with a little downhill tuning. Clean, correct digit — done.”

28 Lesson 28: Does it actually work? A 10-digit stress test

It is fair to ask whether we simply got lucky on one digit. To check, we ran the *exact same method* on ten brand-new digits and watched whether it recovered each one.

The honest setup. MNIST ships as two separate piles: 60,000 *training* images and 10,000 *test* images the method never sees. We used the training pile as the dictionary (the prior), then took one *test* digit of each kind, 0 through 9, simulated its readings with F , and asked the method to recover it. Because the test digits are not in the dictionary, the method cannot cheat by matching a digit to a copy of itself; it must identify the *kind* of digit from the readings alone.

The result: 9 out of 10. Every digit came back correct except one, where an unseen 5 was mistaken for a 6. That single miss is worth understanding, because it *confirms* the lessons rather than undermining them:

- **It was the least-confident case.** Recall from Lesson 8 that the data-match error is really a confidence score. The mistaken 5 had the *worst* Stage-1 error of all ten (about 6×10^{-5} , versus roughly $1\text{--}3 \times 10^{-5}$ for the rest). The method was already “unsure” there, exactly as the theory predicts.
- **That handwritten 5 genuinely resembles a 6.** Its lower half curls into a near-closed loop. With only part of the training set in play, the closest *reading* happened to come from a 6. This is the shadow problem of Lesson 5 made concrete: two different shapes casting almost the same shadow.

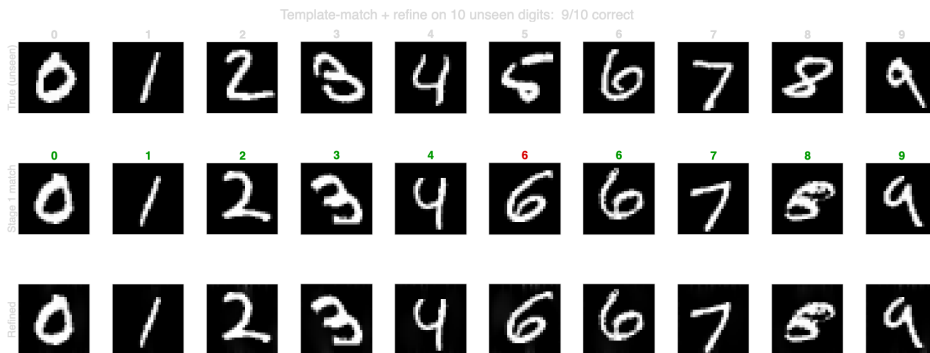


Figure 13: Ten unseen digits put through the method. **Top:** the true (unseen) digit. **Middle:** the dictionary template Stage 1 picked — the number above each is the digit it identified, green when correct, red when wrong. **Bottom:** the refined reconstruction. Nine of ten are correct.

- **The competition 5 is pinned down, not a coin-flip.** On the real data the *top eight* matches were seven 5’s and a single 8 — the leading candidates overwhelmingly agree, which is the signal that the digit is truly identified. The unseen-5 miss was the opposite: a near-tie a 6 won by a hair. Agreement among the top matches is what tells us the competition answer is solid.

Why this matters for the talk. If someone asks “how do you know your 5 is right and not a fluke?”, this is the answer: the same method recovers 9 of 10 *unseen* digits, and on the competition data the top matches nearly all agree on 5. The lone miss is the method honestly flagging its own uncertainty — which is exactly what a trustworthy method should do.

Part VII — Wrap-up

29 Lesson 29: The whole story, start to finish

1. We want the hidden conductivity image from edge voltage readings (Lessons 1–4).
2. It is *ill-posed*: many images fit, and noise blows up — so matching the readings is not enough (Lessons 5–6).
3. We turn “recover the image” into “minimize (data mismatch) + (prior penalty),” where the data part comes from bell-curve noise and the prior part encodes what an answer should look like; a regularizer *is* a prior (Lessons 7–11).
4. A computer minimizes this by walking downhill, using the gradient $J^\top r$ with momentum (Lessons 12–14).

5. Generic priors (small / sparse / few-edges) give blobs; a learned *flat* prior (PCA) gives a clean but *wrong* digit; a learned *curved* prior (VAE) gives the *right* digit — teaching us that prior quality, not data-match error, is what matters (Lessons 15–22).
6. The strongest prior is “the answer is one of the 60,000 real images.” Under it, the optimal method is to *try them all* (template matching) — a global search that nails the digit without getting trapped — and then *polish* the winner. That cleanly recovers the digit (Lessons 23–27).

30 Lesson 30: How this connects to Project 2 (diffusion)

Our winning prior is powerful but rigid: it can only return one of the 60,000 images (plus a little polish). It also returns a *single* answer, even though we saw the data can genuinely be consistent with more than one digit.

Project 2 upgrades the prior with a **diffusion model** — think of it as a vastly more capable version of the VAE’s “curved surface of digits,” able to generate brand-new realistic digits, not just reproduce stored ones. Better still, it can produce *several* different plausible answers (sampling the “many valleys” honestly, instead of forcing one). And the way it folds in the measurements is by using our exact same downhill signal $J^\top r$ at each step. So Project 2 is the natural sequel: same forward simulator, same data-match gradient, but a smarter, generative prior and a fairer way to handle multiple possible answers.

31 Lesson 31: A suggested way to present this

A clean 10–12 minute talk:

1. **Hook:** “We recovered a handwritten digit from 1900 voltage readings around the edge of a box.” Show the final digit.
2. **Setup:** forward (easy) vs. inverse (hard); one image $\leftrightarrow$ 1900 readings.
3. **Why hard:** ill-posed — many images fit, noise blows up (the shadow analogy lands well).
4. **The framework:** minimize (data match) + (prior); and the punch line that *a regularizer is just a prior*.
5. **What we tried:** generic $\rightarrow$ blob; PCA $\rightarrow$ clean but wrong digit; VAE $\rightarrow$ right digit. Deliver the paradox: *lowest error was the wrong answer* (Figure 9).
6. **The villain:** multimodality — many valleys, downhill gets stuck (Figure 7).
7. **The fix:** strongest prior + global search = template matching (show the top-8 agreement, Figure 11); then polish (Figure 12).
8. **Takeaway:** prior quality beats error-chasing; and a one-line tease of Project 2.

Five questions you will probably get — and short answers.

- *Isn't "try all 60k" just cheating/lookup?* No — it is the provably optimal estimate under the stated prior, and Stage 2 refines beyond any stored image. The “lookup” is a principled *global* search that the downhill methods cannot do.
- *Why not approximate each template to go faster?* The digits are high-contrast, so the fast approximation is inaccurate and even picked the wrong digit. Exact simulation is cheap enough.
- *What if the true digit isn't in the 60k?* Stage 1 still finds the closest real digit (identity confirmed by the top-8 agreement), and Stage 2 moves off the template to fit the exact shape.
- *Why did the lowest-error (PCA) result look wrong?* Below the noise level, lower error just means chasing noise; a loose prior exploits that with a wrong image. Error is not correctness.
- *Where does $J^\top r$ come from?* It is the downhill direction of the data term — a vote where each reading pushes on the pixels it affects, weighted by how wrong that reading currently is. It is computed efficiently by the adjoint trick, already inside the simulator.